# Table of Contents

# The Importance of Safety In Our Team

Safety is an important aspect of FIRST as constructing a hundred-pound robot using various industry grade tools and parts can be extremely dangerous. Safety measures are kept in place to minimize injuries, ensure that people know what to do when something goes wrong and to create a good working environment. Our team strongly believes that safety is one of the most important aspects of FIRST. To encourage safety in our team, we have lots of resources that are made known to our team members to make it clear that safety is our number one priority. During competition, our safety captain and our mentors make sure that everyone is following the correct safety procedures established by FIRST. Safety plays an extremely important role in our team's path to success.

# Safety Training

- Our team is safety trained during preseason both e perienced and roo ie e bers are re uired to undergo safety related training and e perienced e bers e en help train and e power new e bers for proper tool use.

- Every member is required to take and pass a hands-on safety test and tutorials. We ha e practices in place to ha e any procedure and new use of tools to be collaborati e.

- All e bers ha e access to our tool/shop guides in our ac of arts oogle dri e and inside our school resources.

- We document the safety training status of each member on our team. Passing and doing safety training is a requirement to be a part of our team. We do re iew this and ha e students tal through and de onstrate proper tool use as needed. We stri e to ha e e ery e ber be able to help with safety related issues and tool operations.

- Our tea understands youth protection plans and has access to all first and a e Washington School District plans.

# Mentor Safety Protocol for contacting families

# Shop and Tool Safety

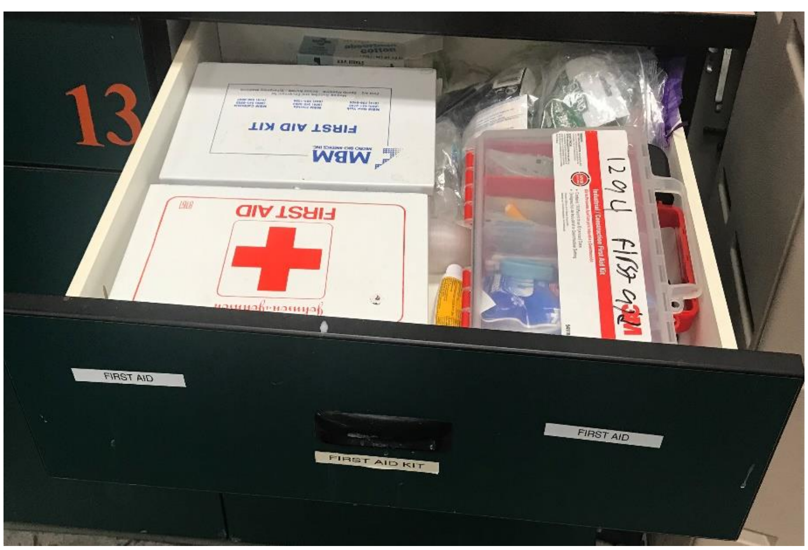
Proper care when using tools in the shop is essential in order to minimize the risk of injury to yourself and other people in the shop. The following rules listed are some of the most important steps to ensure safety in our shop:

- **Report all injuries or accidents to a mentor.**
- Always wear safety glasses when you **operating or near power tools, and other dangerous equipment.**
- Make sure to wear ear protection when operating **and near** loud machinery.
- If your hair is long **or are wearing loose fitting clothing**, make sure to tie it back **or secure properly.**
- Make sure that your hands are dry when operating electrical devices.
- Always use the proper tool for the job. **Ask when unsure**
- Make sure that **all parts of the** tool is in good condition before using it.
- Make sure to know how to use a tool before using it. **Check team information and signage on machines.**
- Make sure to use a clamp to secure the material that you are using before cutting. **Ask for help as needed**
- Don't get distracted when operating power equipment.
- Hold tools power tools firmly when you operate them.
- Direct cutting strokes away from your body and be aware of your surroundings. **Sparks, dust, fumes will impact those next to you.**
- Store sharp-edged tools **should have shield to protect.**

- Never leave a machine unattended while it is running.
- Make sure that the cutting tool is properly installed on the machine before operating it.
- **Use dust collection and proper ventilation whenever possible.**
- If a machine isn't working properly or doesn't sound right, turn it off and tell a mentor.
- Use a piece of scrap wood to move scraps around the cutting tool; **use brooms, and vaccums whenever possible.**
- Make sure to use wood blades to cut wood and metal blades to cut metal.
- Make sure to disconnect the power tool if it's not in use.
- Horseplay is never allowed inside the shop.
- **Be aware of your surroundings at all times.**
- **Eye wash station is located next to sink in D125, or water bottle can be sprayed into eyes as needed.**

## First Aid

- Our team has a dedicated First Aid drawer to store First Aid supplies located next to the sink. and Drawers 13. Orange first aid kits is always brought with the competions.
- UL & FIRST Safety Learning Portal (learnshare.com) is a safety training to explore
- Safety | FIRST (firstinspires.org)
- If you get an injury at any time, be sure to let a mentor know so that they can assist you.



# CNC Safety

- Always wear safety goggles while operating the CNC.
- Wear ear protection to protect your hearing as the CNC is very loud.
- Make sure that you know where the E-Stops are and when to use them. If you do not know where they are located or what they are, do not operate the CNC and request someone else to operate it.
- If you have not used the CNC before, study the CNC user guide found on the OneNote, and then ask someone experienced with the CNC to show you the process. DO NOT "wing it" or proceed without proper instruction or knowledge of the CNC as you will hurt yourself and those around you while damaging the CNC.
- Team members have access to a complete CNC user guide via the team documentation hub, OneNote.
- Make sure everyone near the CNC is wearing safety goggles and ear protection and inform someone before you start operating it.
- Ensure that the CAD/CAM model has tabs inserted into it to ensure that the piece getting cut does not fly out and harm others or yourself.

- Know when to hit the E-Stops and do not hit one unnecessarily as it can waste stock.

- Make sure that the correct bit is in place as the incorrect one can break either the stock or the bit itself and can harm people.

- Ensure that the feeds and speeds withing the CAD/CAM are appropriate for the stock and bit you are using (you will have to adjust the slider on top of the spindle to adjust speed of the bit).

- Always have a spotter with you to help keep you safe (you have to ensure their safety as well).

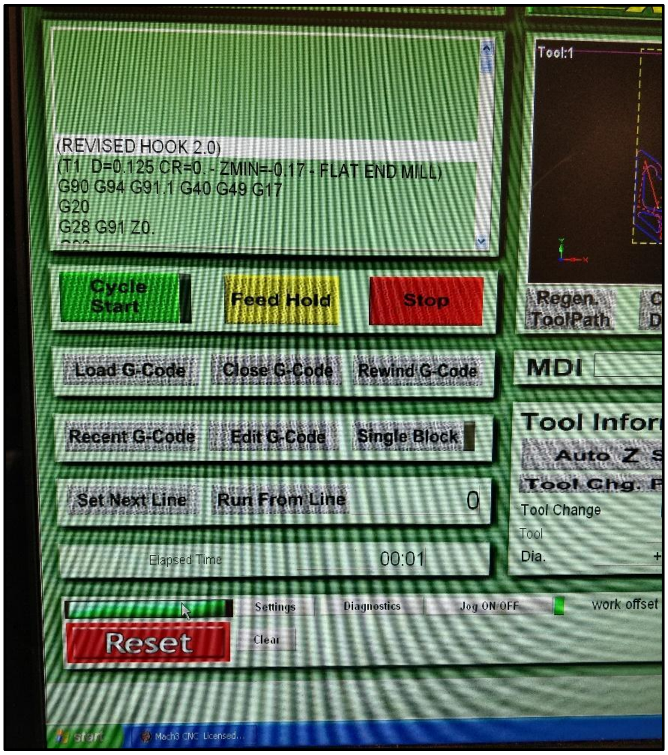
## What If Something Goes Wrong?

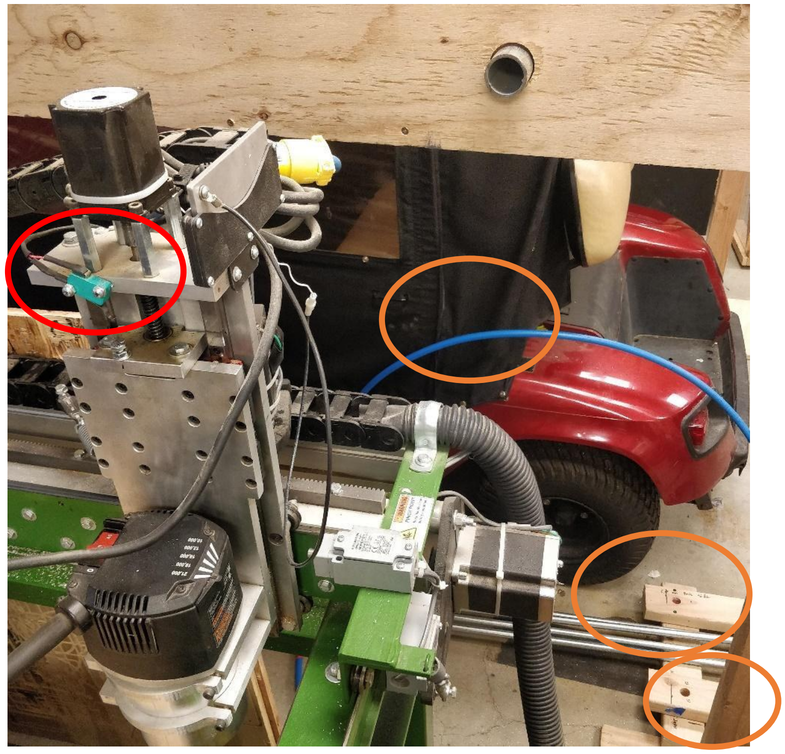
- If something has gone wrong or the CNC does not work, do not attempt to fix the problem yourself. Call a mentor or someone who has more experience on the CNC than you. Discuss what went wrong with them (more than 1 person) and fix the problem.

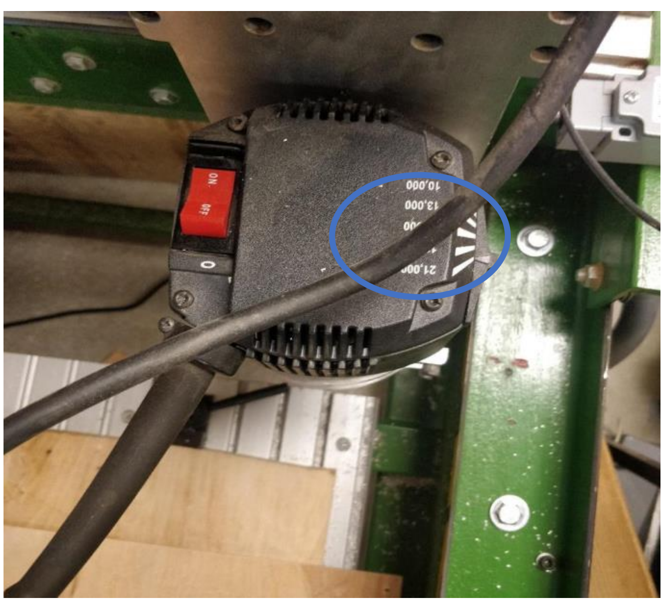
- Do not simply restart the process, go back to CAD/CAM and check over everything such as max stepdown and heights. Also check if it was your setting of the zeroes that caused the error.

- Every time an error occurs, and you fix it, update the CNC log found on the desktop next to the CNC so that people who use it later on know how to fix the problem. Also inform any frequent CNC users why the problem occurred so that they can prevent it in the future.

# CNC Diagrams







Key:

E-Stops

Spindle speed slider

Limit Switches

# Laser Cutter Safety

- Keep a Fire Extinguisher nearby as a precaution when using the laser cutter.
- Don’t look directly at the laser, as it can potentially cause damage to your eye.
- Always stay attentive and make sure that the laser cutter is performing it’s task correctly.
- If something major goes wrong, press “Stop” or disconnect the laser cutter from power.
- Keep the surrounding area clean.
- Ensure that the vents are removing the smoke that is released.
- Safety glasses while using the laser cutter is not necessary as our laser cutter is covered.

# 3D Printer Safety

- Be aware of toxic fumes from the PLA. Do your best to not stay around the 3D Printers while they're printing for long periods of time.
- Don't touch the surface of the 3D printer while it's printing, heating up, or cooling. Temperatures can be between 200 and 300 degrees Celsius and can cause severe burns.
- Some scrapper blades are laced with contaminants. To avoid exposure, scrape away from your fingers and wear safety gloves.

## Battery Storage and Safety

- Be cautious when handling the robot batteries as they contain acid.
- Don't use robot batteries that are visually damaged in any way.
- Keep the battery charging area nice and clean.
- Battery spillage kit is located next to the battery storage area in our shop. Familiar yourselves with the official FRC 2023 safety manual to know how to use it.
- If there is a battery leak at a FRC Competition, go and ask pit administration for help.
- Follow the corresponding procedure to handle a leaking battery (located in the official FRC 2023 safety manual).
- Be careful when lifting the battery as they can be heavy. Hold it straight up to avoid any leaks.
    - This year, our team has developed a 3D printed handle design to go on top of the robot battery. This makes the battery much easier to carry around.

# COVID-19 Safety and general sickness safety

Since the 2020 global COVID-19 pandemic, there has been an emphasized caution on safety procedures to minimize potential spread of the coronavirus. As the world has begun a return to normalcy, the caution has been minimized but it's still important that we do not have a breakout within the team. Below are some of the precautions that we encourage all team members to follow:

- Follow all US/WA DOH and CDC guidelines for containing the spread of all transferable illnesses.

# Sound, eye, air, food safety

Exposure to **Noise** is measured in units of sound pressure levels called decibels, using an A-weighted sound levels (dBA). There are several ways to control and reduce worker exposure to noise in a workplace where exposure has been shown to be excessive.

Engineering controls involve modifying or replacing equipment, or making related physical changes at the noise source or along the transmission path to reduce the noise level at the worker's ear. Examples of inexpensive, effective engineering controls:

Choose low-noise tools and machinery
Maintain and lubricate machinery and equipment (e.g., oil bearings)
Place a barrier between the noise source and employee (e.g., sound walls or curtains)
Enclose or isolate the noise source.

We have members of our team with various **food** allergies and need to be considerate of needs. We also want to be sure we are using proper sanitation methods and be aware that food borne illnesses are possible as we eat at competitions and bring in food from outside sources.

Air safety is dependent on the many materials we use each day for our robot. Some to be aware of, dust collector for any wood related tool use, acrylic cutting on laser emits toxic fumes, some printer filaments need to have ventilation to prevent the inhalation of microplastics, polycarbonate emits fumes that are unpleasant to smell.
"The relative importance of any single source depends on how much of a given pollutant it emits, how hazardous those emissions are, occupant proximity to the emission source, and the ability of the ventilation system (i.e., general or local) to remove the contaminant. In some cases, factors such as the age and maintenance history of the source are significant." (OSHA)

Eyes are one of the most injured organ on the human body. Be sure to use safety glasses whenever possible. We have eye wash stations and have first aid protocols for helping with proper bandages.

# INFECTION CONTROL PLAN

## Purpose:

Lake Washington School District #414 is committed to providing a safe and healthful work environment for our entire staff. In accordance with the WISHA Bloodborne pathogens standard WAC 296-823, Lake Washington School District has developed the following exposure control plan to reduce employee occupational exposure to blood or other potentially infectious materials as detailed in the Bloodborne Pathogens standard.
Employees who have occupational exposure to blood or other potentially infectious material (OPIM) must follow the procedures and work practices in this plan.
Employees can review this plan at anytime during their work shifts. We will provide a copy, free of charge, to an employee within 15 days of a request.

## Administration and Compliance:

The Risk Manager is the administrator of this plan and is responsible for its implementation.

Employees who are identified as having occupational exposure are required to comply with the procedures and work practices outlined in this exposure control plan. Failure to follow these procedures can result in disciplinary action.

## Exposure Determination:

WISHA requires employers to perform an exposure determination to identify employees who have occupational exposure to blood or other potentially infectious materials. Occupational exposure means “reasonably anticipated skin, eye, mucous membrane or parental contact with blood or other potentially infectious material that may result from the performance of an employee’s duties.” Employees are considered to have occupational exposure even if they utilize PPE while performing duties that put them at risk for exposure. For purposes of determining which employees are considered to have occupational exposure. Listed below are the job classifications and tasks in which employees may be expected to incur such occupational exposure, regardless of frequency.

| Job Classifications in which all employees have occupational exposure | Tasks With Exposure |
|---|---|
| Custodians | Spill clean-up of body fluids, disposal of hazardous materials, restroom maintenance |
| Nurses<br>Health Room Office Professionals | First aid, medical procedures, blood glucose monitoring, intermittent catheterization, gastrostomy feedings, oral suctioning, toileting procedures |
| Para-Educators | Unpredictable behavior with potential body fluid exposure, toileting procedures, cleaning intermittent catheterization, gastrostomy feedings, oral suctioning |
| Pre-School Teachers (with playground supervision responsibilities) | First aid on playground |
| Instructional Assistants | First aid on playground |
| School Building Office Staff | Responsible for health room when Nurse is not available, first aid |
| Coaches/Athletic Trainers | Responders for injuries during athletic events, first aid, injury management |
| **Job Classifications in which some employees have occupational exposure** | **Tasks With Exposure** |
| Bus Drivers | First aid, unpredictable behavior with potential body fluid exposure, spill clean up |
| OT/PT | Toileting procedures, unpredictable behavior with potential for body fluid exposure |
| Security | Student behavior management, first aid |

## Contact names and phone numbers:

- *Risk Management (425)936-1119* will maintain, review, and update the exposure control plan at least annually, and whenever necessary to include new or modified tasks and procedures.
- *Risk Management (425)936-1119* will make this plan available to employees, and WISHA (Washington Industrial Health and Safety Act)

## Universal Precautions:
Universal precautions are an approach to infection control. According to the concept of universal precautions, all human blood and certain human body fluids are treated as if known to be infectious for bloodborne pathogens. Universal precautions will be observed throughout the Lake Washington School District in order to prevent contact with blood or other potentially infectious materials.
*Note*:
Universal Blood-Body Fluid Precautions, Body Substance Isolation, and Standard Precautions expand on the concept of universal precautions to include all body fluids and substances as infectious. These concepts are acceptable alternatives to universal precautions.

## Exposure Determination Annual Review:
The Infection Control Plan is reviewed (and if necessary up-dated) annually by the Lake Washington Infection Control Committee and in addition, whenever necessary to reflect new or modified tasks and procedures which affect occupational exposure and to reflect new or revised employee positions with occupational exposure. This committee will meet formally annually, convened by the Risk Manager.

## Engineering Controls:
Lake Washington School District conducts ongoing evaluation of tasks and medical devices that carry a risk of exposure and implements safer medical devices whenever feasible.

We have developed the following engineering controls to prevent or minimize exposure to bloodborne pathogens. New technology will be implemented and evaluated whenever possible. Our engineering controls will be evaluated and maintained as described in the following:

| Controls In Use | Location | Evaluation/Service Interval | Controls Evaluated |
|---|---|---|---|
| Sharps container | In all health rooms | When ¾ full | Annually |

## Work Practice Controls:
The following work rules apply where there is potential for contact with blood or OPIM:

### Hand Washing
Hand washing facilities are available to employees who are exposed to blood or other potentially infectious materials.

- Employees shall wash hands after removal of personal protective gloves and whenever there is a likelihood of contamination. In addition, any contaminated skin area will be washed as soon as possible.
- When hand washing facilities are not readily available, the use of waterless hand washing products is permitted as an interim means of washing the hands or other parts of the body after contamination with blood or OIPM.
- When it is not possible to provide hand washing facilities, an appropriate hand cleanser (containing alcohol) shall be used in conjunction with clean cloth/paper towels or antiseptic towelettes. When antiseptic hand cleaners or towelettes are used, hands shall be washed with soap and running water as soon as possible.

- **It is our strict policy that employees wash their hands immediately or as soon as possible after removal of gloves or other personal protective equipment.**
- This policy also requires that employees wash hands or any other skin with soap and water or flush mucous membranes with water immediately or as soon as possible following contact with blood or other potentially infectious materials.

**Sharps Containers**
Sharps containers are supplied in each health room.

- Contaminated needles may not be recapped, bent or broken off. Shearing or breaking of contaminated needles is prohibited.
- Contaminated sharps must be deposited in a sharps container immediately or as soon as possible after use. These containers shall be puncture resistant, with orange or orange-red labels with contrasting lettering or symbols, and leak proof on the sides and bottom.
- Reusable sharps (including lances for diabetic glucose monitoring) that are contaminated with blood or other potentially infectious materials shall not be stored or processed in a manner that requires employees to reach by hand into the containers where these sharps have been placed.
- All labels shall have the biohazard legend.
- Sharps containers must be closed prior to removal or replacement to prevent spilling or protrusion of the contents during handling or storage.
- Link to Sharps Containers Guidelines:

**Other work practices**

- Eating, drinking, applying cosmetics or lip balm, and handling contact lenses are prohibited in the work areas where there is reasonable likelihood of occupational exposure.
- Food and drink must not be kept in refrigerators, freezers, shelves, and cabinets or on countertops or bench tops where blood or other potentially infectious materials are present.
- All procedures will be conducted in a manner that will minimize splashing, spraying, splattering, and generation of droplets of blood or other potentially infectious materials.
- Equipment which may be contaminated with blood or infectious materials must be examined prior to service or shipping and shall be decontaminated as necessary. If decontamination is not feasible then a readily observable biohazard label shall be attached to the equipment and the contaminated portions documented.
- Employees shall observe universal precautions and utilize appropriate PPE when handling such equipment.

## Personal Protective Equipment (PPE)

When there is an exposure to bloodborne pathogens or other potentially infectious material, the Lake Washington School District shall provide, at no cost to the employee, appropriate personal protective equipment based on the anticipated exposure to blood or other potentially infectious materials.

The PPE will be considered appropriate only if it does not permit blood or other potentially infectious materials to pass through or reach the employees' clothing, skin, eyes, mouth, or other mucous membranes under normal conditions of use and for the duration of time that the protective equipment will be used.

Employees will receive training on the appropriate use of PPE provided for specific tasks. The following personal protective equipment is provided for workers:

| PPE | Use Guidelines |
|---|---|
| Disposable Gloves | Anticipated body fluid exposure |
| Utility Gloves | Disposing of contaminated waste |
| Safety Glasses w/side shields | Unpredictable behavior with potential for body fluid exposure, toileting, suctioning |
| Face Shield | Unpredictable behavior with potential for body fluid exposure, toileting, suctioning |

| | |
|---|---|
| Apron | Spill clean-up |
| CPR Shield | Emergency procedures |
| Neoprene gloves | Unpredictable behavior with potential for body fluid exposure |

Refer to department procedures for instructions on the use of PPE for specific tasks that may expose workers to blood or other potentially infectious material.

If required PPE is not available, contact the building administrator who will insure that supplies are replaced.

### Gloves

- Gloves shall be worn where it is reasonable to anticipate that employees will have hand contact with blood, other potentially infectious material, non-intact skin, and mucous membranes.
- Disposable (single use) gloves shall not be washed or decontaminated for re-use and are to be replaced as soon as practical when they become contaminated or as soon as possible if they are torn, punctured, or when their ability to function as a barrier is compromised.
- Utility gloves may be decontaminated for re-use provided that the integrity of the glove is not compromised. Utility gloves will be discarded if they are cracked, peeling, torn, punctured, or exhibit other signs or deterioration or when their ability to function as a barrier is compromised.

### Other PPE

- Appropriate face and eye protection must be worn when splashes, sprays, splatters, or droplets of blood or other potentially infectious materials pose a hazard to the eyes, nose or mouth.
- Gowns and other protective body clothing shall be worn whenever there is a risk of splash to the body.
- All garments that are penetrated by blood shall be removed immediately or as soon as possible. All personal protective equipment will be removed and placed in a designated area or container prior to leaving the work area.

### Accessibility

The Lake Washington School District shall ensure that appropriately sized personal protective equipment is readily accessible or issued to our employees.

### Cleaning, Laundering, and Disposal

The Lake Washington School District repairs and replaces personal protective equipment as needed to maintain its effectiveness, at no cost to employees. Garments penetrated by blood or other potentially infectious materials shall be removed immediately or as soon as possible.
In most circumstances, employees shall remove personal protective equipment prior to leaving the work area. When personal protective equipment is removed, it shall be placed in an appropriately designated area or container for disposal.

## Housekeeping:

All equipment and work surfaces shall be cleaned and decontaminated as soon as possible after contamination with blood or OPIM and at the end of the work shift if the surface may have become contaminated since the last cleaning. Broken glassware which may be contaminated with blood or other potentially infectious material shall never be stored or processed in a manner that requires an employee to reach by hand into the containers used for their disposal.

## Handling of Regulated Waste Material:

Regulated waste is defined as liquid or semi-liquid or other potentially infectious materials, contaminated items that would release blood or other potentially infectious materials in a liquid or semi-liquid state is compressed, items that are caked with dried blood or other potentially infectious materials and are capable of releasing these materials during handling, contaminated sharps, and pathological and micro-biological waste containing blood or other potentially infectious material.

Contaminated Sharps Discarding and Containment (Sharps Containers Guidelines link):

- Contaminated sharps shall be discarded immediately or as soon as possible into containers that are: closable, puncture resistant, leak proof on sides and bottom, and labeled using orange or orange-red labels with lettering or symbols in a contrasting color or red in color.
- During use, containers for contaminated sharps shall be:
    - Easily accessible to personnel and located as close as possible to the area where sharps are used or can be reasonably anticipated to be found.
    - Maintained upright throughout use, and
    - Replaced routinely and are not allowed to be over filled.
- When moving containers of contaminated sharps in order to prevent spillage or protrusion of contents, the container shall be:
    - Closed immediately prior to removal or replacement, and
    - Placed in a secondary container if leakage is possible.
- When a sharps container is ¾ full, the top is to be closed and taped securely in an "X" design. Transportation of the container is arranged by using the Transfer of Equipment form available in public folders.
- Never manually open, empty, or clean reusable contaminated sharps disposal containers. They must be cleaned according to the manufacturer's instructions.
- Other waste shall be placed in a labeled biohazard waste container in each nurse's office and specific classrooms. This waste will be tied off separately and disposed of in the regular garbage can.
- Remove and replace protective coverings such as plastic wrap and foil on equipment and surface when they become contaminated.
- Always use mechanical means such as tongs, forceps, or a brush and dustpan to pick up contaminated broken glass. **Never pick up with hands – even if gloves are worn!**

Signs and Labels:
Warning labels will be placed on containers of waste. The label will be a fluorescent orange or orange-red biohazard label as illustrated with lettering in a contrasting color.

## Hepatitis B Vaccination and Post-Exposure Evaluation and Follow-up:

The Lake Washington School District makes available the Hepatitis B vaccine and vaccination series to all employees who have occupational exposure. The employee will be referred to a competent medical provider currently contracted by the District. The District also provides post-exposure evaluation and follow-up to all employees who have had an exposure incident through workers' compensation.

All Hepatitis B vaccine and vaccination series are:

- Made available at no cost to employees,
- At a reasonable time and place,
- Performed by or under the supervision of a physician or licensed health care professional, and
- Provided according to recommendations of the U.S. Public Health Service.
- If a routine booster dose(s) of Hepatitis B vaccine is recommended by the US Public Health Service at a future date, such booster dose(s) shall be made available at no cost to the employee.
- All laboratory tests are conducted by an accredited laboratory at no cost to the employee.

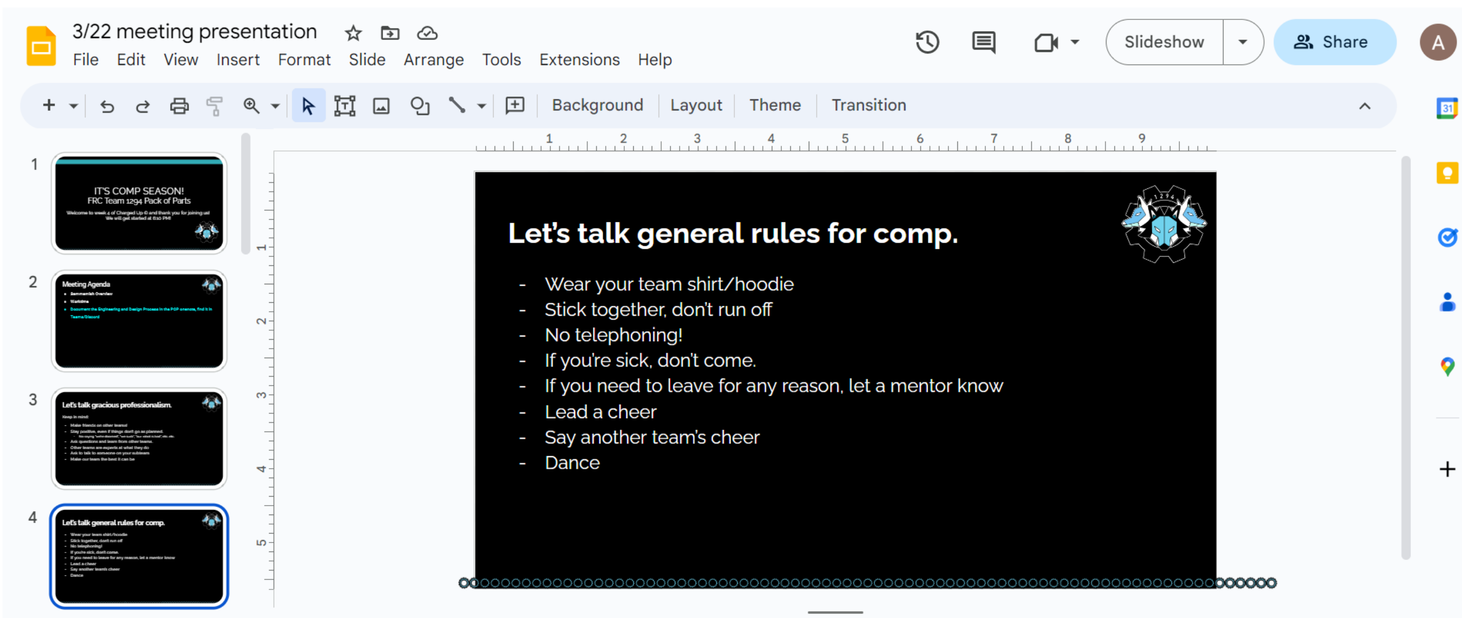
### Hepatitis B Vaccination

All employees who have been identified as having exposure to blood or OPIM through the exposure determination described in this section will be offered the Hepatitis B vaccine series at no cost to the employee within 10 days of initial assignment unless:

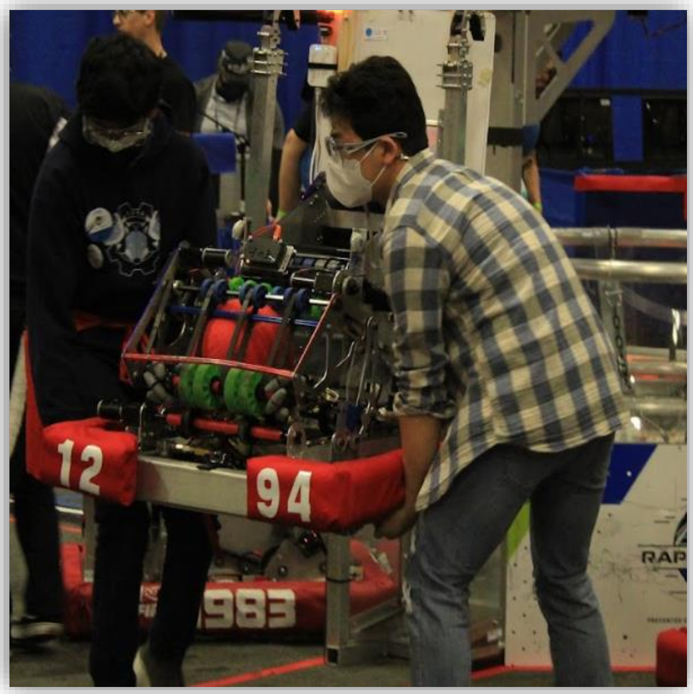
- The employee has previously received the series
- Antibody testing reveals that the employee is immune
- Medical reasons prevent taking the vaccination; or

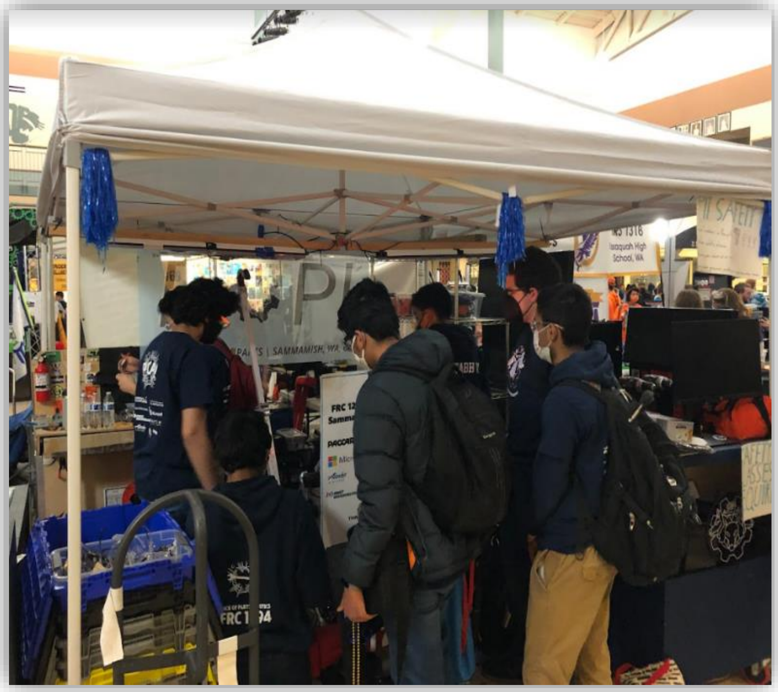
# Competition Safety

Before our first competition, our team always holds a meeting to talk about proper competition safety and behavior. Below are some of the rules and procedures that our members are required to follow:




- Make sure to wear safety glasses when you are in the pits.
- Make sure the robot is on a safe surface when you're working on the robots in the pits.
- Use the buddy system when at comp. Don't wander around by yourself.
- Lift the robot safely in the pits and on the playing field.





- Always keep the pits nice and clean and make sure pedestrians around you are aware of where the robot is at all times.
- Always demonstrate Gracious Professionalism when you are at competition. Offer help to teams who need it and make sure that other teams are following proper safety precautions.
- Make sure that the pit and any other things hung in the pit are secure (cannot be 10ft above the floor).
- Make sure to properly clean up the pits when things have to be taken down.
- Keep a clear path when transporting the robot.
- Make sure to keep track of your belongings.
- Don't throw objects from the stands.



FIRST

- Assist other teams with safety issues.

- Control access to the pits. Make sure that too many people from your team are not at the pits at any time.

- Have proper storage for batteries and tools in the pits.

- Always cooperate with volunteers and judges. If they tell you to do something, do it!

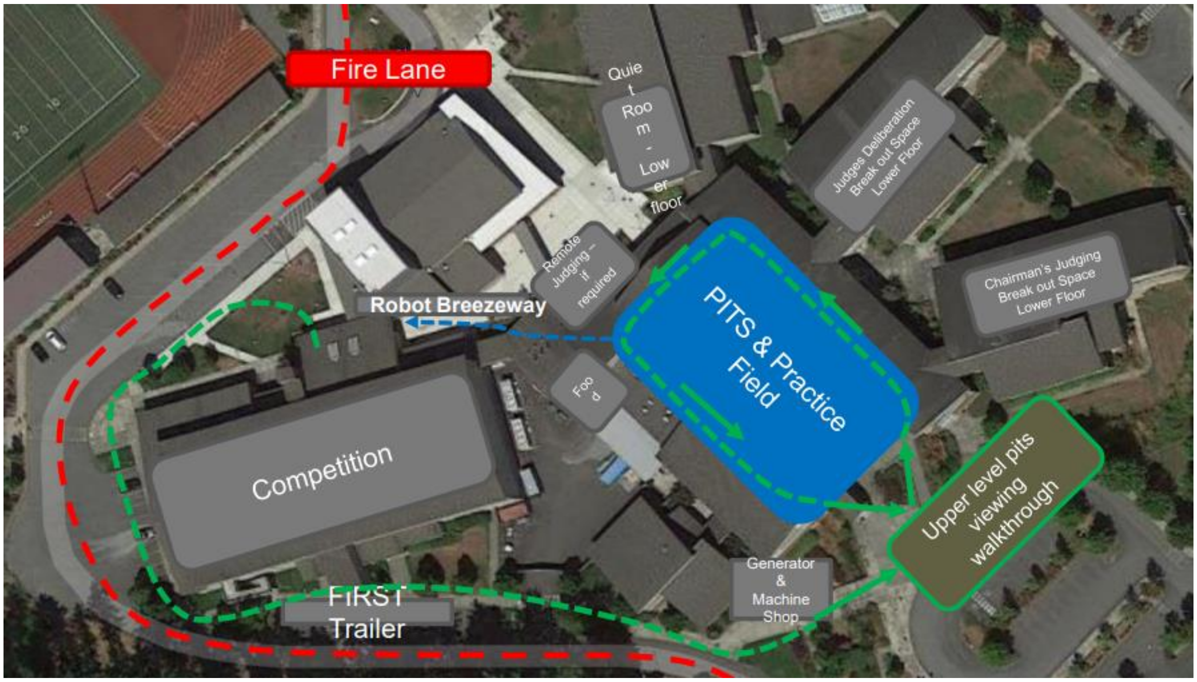
- Report injuries to the EMT (located near the Pit Administration Desk).

- If you have any questions or concerns at safety during competition, talk with the Safety Managers.

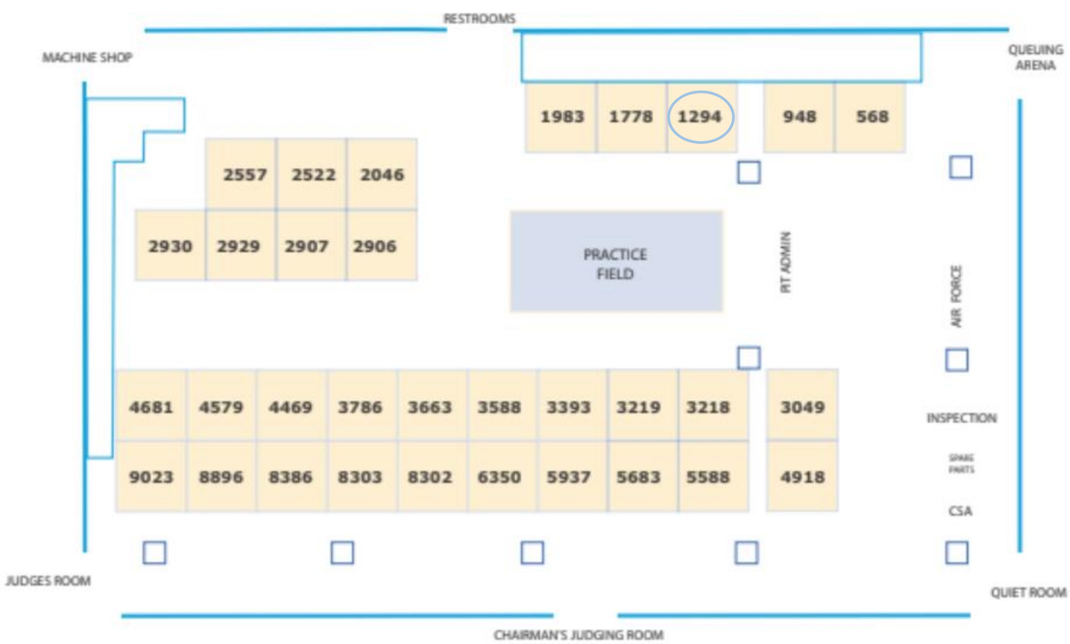
- Be aware of other teams/people in the pits.


# Aub High School FRC Event

## Map:



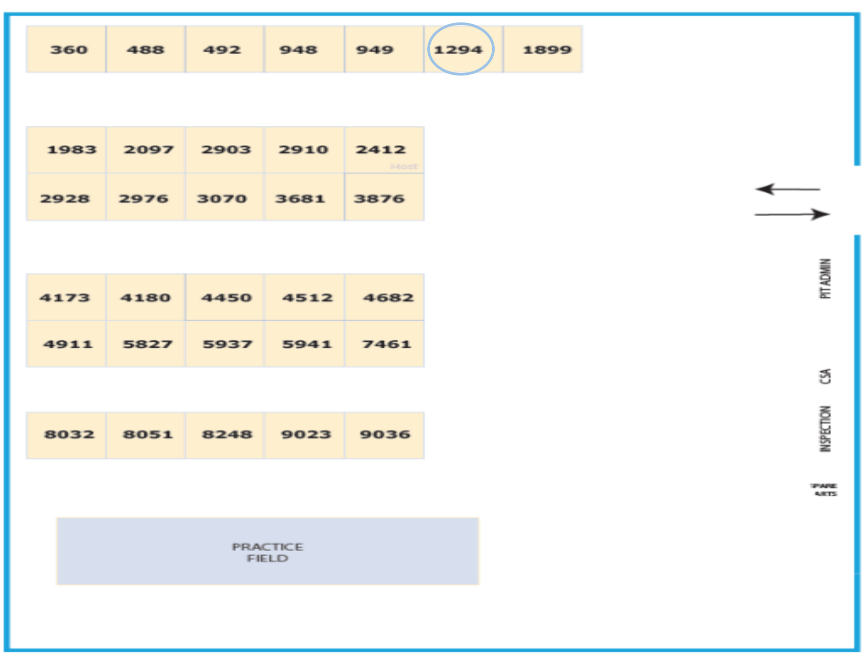

## Pits:



# Sammamish High School FRC Event

## Map:


## Pits:



# Portland OR, DCMP safety

# Houston Texas, World Championship safety

# Outreach event Safety considerations

# Extra Resources

- The Official 2023 FIRST Robotics Competition Safety Manual https://www.firstinspires.org/sites/default/files/uploads/resource_library/frc/team-resources/safety/2023/2023-Safety-Manual.pdf
- Tips protect yourself and others from COVID-19 https://www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/prevention.html
- The World Health Organization's Q&A page for different topics related to COVID-19 https://www.who.int/emergencies/diseases/novel-coronavirus-2019/question-and-answers-hub
- Our Team's Safety Captain: Aditya Manivel.
    - Contact: aditya.manivel@team1294.org